

LESSON 8-1 • SESSION 2 • PRACTICE • FORM B

Writing Equations

Learning target – I can write an equation for a situation and test a proposed answer. · 6.EE.B.6

REMEMBER

- Name the **unknown** with a letter and say what it stands for.
- More / added:** +. **Left / fewer:** −. **Each / equal groups:** ×.
- Estimate, **test** your value, then revise until both sides match.

WORKED EXAMPLE

A baker made 64 rolls. Some were sold, and 27 were left. Write an equation and find the rolls sold.

Step 1 Let r = rolls sold. Made − sold = left: $64 - r = 27$.

Step 2 Test $r = 35$: $64 - 35 = 29$. Too many are left.

Step 3 Test $r = 37$: $64 - 37 = 27$. True.

Answer $r = 37$, so 37 rolls were sold.

WORDS equation (*ecuación*) · variable (*variable*) · unknown (*incógnita*)

PRACTICE

show your work · label units

1. Write an equation for each sentence. Use n .

a. A number plus 23 is 50. _____

b. 8 times a number is 72. _____

c. A number minus 11 is 25. _____

d. A number divided by 6 is 5. _____

2. **Coat Drive.** A coat drive collected c coats on Friday. On Saturday it collected 37 more. The two days together made 82 coats.

a. Write an equation using c . _____

b. Find c . Show one test. _____

3. **Reading Minutes.** Zoe read for m minutes on Monday. On Tuesday she read 12 fewer minutes, which was 44 minutes.

Hint Fewer means subtract from Monday's minutes.

a. Write an equation using m . _____

b. Test $m = 50$. Revise to find the solution. _____

4. **Bus Riders.** A bus left school with 52 students. After the first stop, 35 were still on. Write an equation for g , the students who got off. Solve it.

Hint Start minus got off leaves still on.

Show your work.

$g =$ _____

5. **Aquarium Water.** A full tank holds 80 liters. Right now it holds 47 liters. Write an equation for w , the liters still needed. Solve it.

Hint What you have plus what you need fills the tank.

Show your work.

$w =$ _____

6. **Food Drive.** A class collected c cans. They donated 64 cans and had 29 cans left. Write an equation. Then estimate and check to find c .

Hint The cans collected must be more than 64.

Show your work.

Equation; c _____

7. **Team Buses.** A team fills b buses with 40 students each, for 160 students in all.

- a. Write an equation for the number of buses. _____
b. Test $b = 4$. Is it the solution? _____

8. **Field Trip Cost.** A trip costs a \$90 fee plus \$6 for each student. The total bill is \$270.

- a. Define a variable. Write an equation. _____
b. Test 30 students. Does it fit the bill? _____
c. If there were no \$90 fee, what equation fits a \$270 bill? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Priya's Work.** A bag had 60 beads. Some were used, and 18 were left. Priya wrote $18 - b = 60$ and answered $b = 42$. Explain her mistake. Write the correct equation.

TEST PRACTICE

choose the best answer

10. "7 times a number equals 63" is written –

- A $7 + n = 63$ B $n - 7 = 63$ C $7n = 63$ D $n \div 7 = 63$

11. "A number decreased by 6 equals 21" is written –

- A $n + 6 = 21$ B $6 - n = 21$ C $6n = 21$ D $n - 6 = 21$

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

32 Z	45 S	119 Q	93 E	35 J	33 V	87 X	17 L	56 O
		#2b	#3b	#4	#5	#6		

CHECK YOURSELF

- ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____